Lecture 4(a): Introduction to optimization

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Agenda

- Introduction to optimization
- Mathematical formulations
- Examples of optimization models in power system

Intro to optimization

What is an Optimization Problem?

- **Definition of Optimization Problem**: Optimization problems seek the best decision from a set of feasible options defined by constraints.
- **Objective Function Role**: The objective function quantifies performance to maximize efficiency or minimize cost in optimization.
- **Constraints in Optimization**: Constraints define the feasible region by limiting decision variables through equations or inequalities.
- **Types of Optimization Problems**: Optimization includes linear, integer, convex, and nonlinear problems modeling real-world scenarios.

Objective Function and Constraints

- **Role of Objective Function**: The objective function defines the optimization goal, such as minimizing cost or maximizing performance.
- **Influence of Function Type**: Linear objectives simplify solving, whereas nonlinear or nonconvex forms increase problem complexity.
- **Nature of Constraints**: Constraints specify limits like capacity or safety, shaping the set of acceptable solutions.
- **Feasible Region Geometry**: Linear constraints create convex polytopes; nonlinear constraints form curved or disconnected solution spaces.

Importance of Linearity and Convexity

- **Efficiency of Linear Problems**: Linear functions and constraints allow efficient solutions using simplex or interior-point algorithms.
- **Convexity and Global Optimality**: Convex problems guarantee global optimum and have strong duality, aiding solution diagnostics.
- **Challenges of Nonconvexity**: Nonconvex problems can have multiple local optima and are often NP-hard and difficult to solve.
- **Model Reformulation Importance**: Preserving convexity through reformulation is key in modern optimization modeling approaches.

Mathematical formulations

Linear programing

LP involves minimizing a linear objective function with linear equality and inequality constraints defining the feasible region.

Finds the minimum of a problem specified by

$$\min_{x} f^T x \text{ such that } \begin{cases} A \cdot x \leq b, \\ Aeq \cdot x = beq, \\ lb \leq x \leq ub. \end{cases}$$

f, x, b, beq, lb, and ub are vectors, and A and Aeq are matrices.

- Optimization variables in the form of a vector x.

Mixed integer linear programming

MILP extends linear programming by allowing integer or binary decision variables alongside continuous ones.

Finds the minimum of a problem specified by

$$\min_x f^T x \text{ subject to } \begin{cases} x(\text{intcon}) \text{ are integers or extended integers} \\ A \cdot x \leq b \\ Aeq \cdot x = beq \\ lb \leq x \leq ub. \end{cases}$$

f, *x*, *b*, *beq*, *lb*, and *ub* are vectors, intcon is a vector or an integer Constraint object, and *A* and *Aeq* are matrices.

Convex optimization

Convex optimization involves minimizing a convex objective function subject to convex inequality and affine equality constraints, ensuring global optimality.

Convex optimization is the mathematical problem of finding a *x* vector that minimizes the function:

Objective function:

$$min_x f(x)$$

Constraint:

$$g_i(x) \leq 0 \text{ (nonlinear inequality constraints)}$$

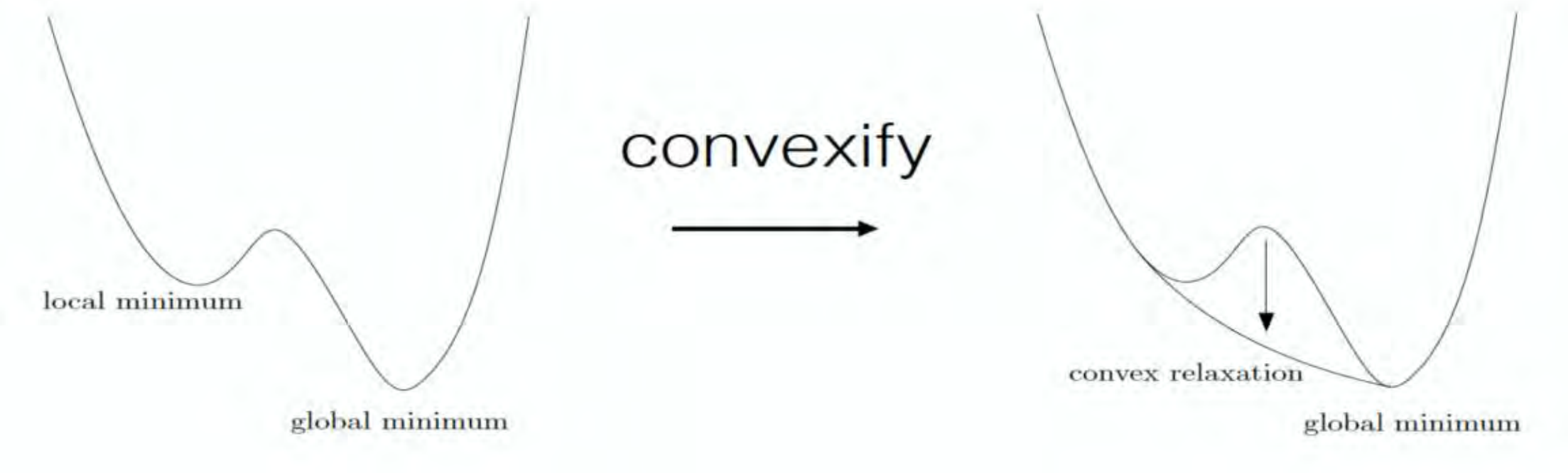
$$Ax \leq b \text{ (linear inequality constraints)}$$

$$A_{eq}x = b_{eq} \text{ (linear equality constraints)}$$

$$lb \leq x \leq ub \text{ (bound constraints)}$$

where $g_i, i = 1, \ldots, m$ are convex functions.

Convexification of non-convex optimization



Computational complexity

P vs NP

$$\underbrace{\text{LP} \subset \text{QP} \subset \text{CQCP} \subset \text{SOCP} \subset \text{SDP}}_{\text{Convex}} \subset \underbrace{\text{NCQCP} \subset \text{NLP}}_{\text{Nonconvex}}$$

- polynomial time
- *nondeterministic* polynomial time
- ≈exponential

n variables	$O(n^k)$	$O(k^n)$
k constant	$O(n^2)$	$O(2^n)$

Optimization solvers

		maturity	mixed-integer?	large-scale?
LP	linear	high	yes	yes
SOCP	convex	high	yes	yes
SDP	convex	initial commercialization	no	research
NLP	(non)convex	depends on the problem (instance)	depends on the problem (instance)	depends on the problem (instance)

Second order cone relaxation (SOC), Semi-definite relaxation (SDP), Quadratic Convex relaxation (QC)

Examples of optimization models in power system

Power system problems

Table 1:Major types of power system problems

General problem type	Problem name	Includes voltage angle constraints?	Includes bus voltage magnitude constraints?	Includes transmission constraints?	Includes losses?	Assumptions	Includes generator costs?	Includes contingency constraints?
OPF	ACOPF, or Full ACOPF	Yes	Yes	Yes	Yes		Yes	No
OPF	DCOPF	No	No; all voltage magnitudes fixed	Yes	Maybe	Voltage magnitudes are constant	Yes	No
OPF	Decoupled OPF	Yes	Yes	Yes	Yes	Power-voltage angle are independent of voltage magnitude-reactive power	Yes	No
OPF	Security-Constrained Economic Dispatch (SCED)	Yes	No	Yes	Yes	Voltage magnitudes are constant	Yes	Yes
Power flow	Power Flow, or Load Flow	No, but can be added	Yes	No, but can be added	Yes		No	No
Economic dispatch	Economic Dispatch	No	No	No	Depends	No transmission constraints	Yes	No
OPF	Security Constrained OPF (SCOPF)	Yes	Depends	Yes	Yes	Depends	Yes	Yes

Source: Resources about AC-OPF from the US Federal Energy Regulatory Commission (FERC), Optimal Power Flow and Formulation Papers | Federal Energy Regulatory Commission

Complexity of power system problems

Problem	Type	Complexity
Economic dispatch	LP	P
DC OPF	LP	P
AC OPF (Non-convex)	NLP	NP-hard
Unit commitment	MILP	NP-hard
Optimal planning	MILNP	NP-hard

Any questions?

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